



Reading: DMAIC Cheat Sheet

Estimated duration: 3 minutes

Introduction

The DMAIC framework is the backbone of Six Sigma improvement projects. It provides a structured, data-driven approach for solving process problems and sustaining results. Each phase—Define, Measure, Analyze, Improve, and Control—builds upon the previous one to ensure decisions are based on evidence, not assumptions. DMAIC applies across industries such as manufacturing, logistics, healthcare, finance, and services. This reading helps you understand how DMAIC tools improve and sustain process performance.

DMAIC quick reference cheat sheet

Phase	Purpose	Key Activities and Tools	Stakeholders
Define	Clarify the problem and align with business goals.	- Develop Project Charter - Identify VOC & CTQ needs - Create SIPOC diagram - Set boundaries and objectives	Project Sponsor/Champion, Black Belt/Green Belt, Process Owner, Key Stakeholders
Measure	Understand the current state and establish baseline performance.	- Map the process - Develop data collection plan - Gather reliable data - Validate measurement system	Belts, Analysts, SMEs, Data Owners / IT
Analyze	Discover root causes of variation, errors, or waste.	- Root Cause Analysis (5 Whys, Fishbone, Pareto) - Use statistical tests - Identify VA vs. NVA steps - Prioritize causes by impact	Belts, Analysts, SMEs, Team Members
Improve	Design, test, and implement solutions that address root causes.	- Brainstorm and pilot ideas - Apply Lean tools - Pilot testing - Measure impact	Project Team, Process Owner, Supervisors/Operators, Leadership
Control	Sustain the gains and ensure long-term stability.	- Create Control Plan - Update SOPs & training - Assign ownership - Monitor using KPIs	Process Owner, Supervisors/Quality, Belts, Governance Committee

Avoiding pitfalls in DMAIC

Common Pitfalls	Steps to Avoid Pitfalls
Jumping to solutions early	Validate the problem and align stakeholders first.
Poor or incomplete data	Create a strong measurement plan and verify accuracy.
Weak root cause analysis	Validate causes with data, not assumptions.
Skipping pilot testing	Test improvements on a small scale first.
Weak control & follow-up	Include dashboards, audits, and ownership.
Excluding frontline employees	Engage those closest to the work.

Summary

The **DMAIC framework**, Define, Measure, Analyze, Improve, and Control, is the core roadmap of Six Sigma. It provides a structured, data-driven approach for improving existing processes and sustaining results.

Each phase builds logically on the one before it:

- **Define** clarifies the problem and aligns stakeholders.
- **Measure** establishes the baseline and gathers reliable data.
- **Analyze** uncovers verified root causes.
- **Improve** designs and tests effective solutions.
- **Control** ensures that gains are sustained through monitoring and accountability.

When used with Lean and systems thinking, DMAIC helps teams make smarter decisions, eliminate waste, reduce variation, and create lasting, measurable improvement across the organization.

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